Question	Answer	Marks	Partial Marks
5(a)(ii)	28 nfww	**4**	**M1** for midpoints **soi** **M1** for use of $\sum fm$ with m in correct interval including both boundaries **M1** (dep on 2nd M1) for $\sum fm \div 80$
5(b)(i)	$\frac{7}{8}$ cao	**2**	**M1** for $\frac{18+28+24}{80}$ oe
5(b)(ii)	$\frac{25}{126}$ oe	**3**	**M2** for [2 ×] $\left(\frac{3}{28}\times\frac{25}{27}\right)$ or [2 ×] $\left(\frac{25}{28}\times\frac{3}{27}\right)$ oe or **M1** for either $\frac{3}{28}$ or $\frac{25}{27}$ or $\frac{25}{28}$ or $\frac{3}{27}$ If 0 scored, **SC1** for answer $\frac{75}{392}$ oe
5(c)(i)	28 and 56	**1**	
5(c)(ii)	Correct diagram	**3**	**B1FT** *their* **(c)(i)** for plots at 5 correct heights **B1** for 5 plots at upper ends of intervals on correct vertical line **B1FT** (dep on at least B1) for increasing curve or polygon through 5 points After 0 scored, **SC1FT** for 4 correct points plotted
5(c)(iii)	Strict FT *their* reading at 80th percentile for an increasing curve/polygon	**2**	**B1** for 64 written or a mark at cf = 64 on graph or a mark on curve at (t, 64)
5(c)(iv)	Correct integer reading at $t = 45$	**M1**	**FT** *their* cf graph for all three marks
	$\frac{80-(\textit{their}\text{ reading at } t=45)}{80}$ [× 100] or $\frac{(\textit{their}\text{ reading at } t=45)}{80}\times 100$	**M1**	
	Percentage consistent with *their* reading	**A1**	If no working shown then SC1 for a correct percentage that follows from a correct reading from *their* graph.
6(a)	$5b - 2a$ final answer	**2**	**B1** for $5b$ or $-2a$ in final answer or for $5b - 2a$ seen
6(b)	$6x - 23$ final answer nfww	**2**	**M1** for $4x - 20$ or $-3 + 2x$

Question	Answer	Marks	Partial Marks
6(c)	$\frac{35-x}{2x(x-5)}$ or $\frac{35-x}{2x^2-10x}$ oe final answer nfww	**3**	**B1** for 3(2x) – 7(x – 5) or better isw **B1** for 2x(x – 5) as common denominator isw, allow expanded
6(d)	–5	**3**	**M1** for 13 – 4x = 18 – 3x oe or $\frac{-4x}{3}+x=6-\frac{13}{3}$ oe **M1FT** for −4x +3x = 18 – 13 oe or for $\frac{-x}{3}=\frac{5}{3}$
6(e)	[x =] $\frac{5p}{y+10}$ oe final answer	**4**	**M1** for correctly clearing the x from the denominator **M1** for correctly expanding the brackets or (dealing with the 5 correctly throughout) **M1** for correctly isolating terms in x **M1** for correctly factorising and dividing by the bracket Max 3 marks if answer is incorrect
7(a)	87.[0] or 86.98 to 86.99	**3**	**M2** for $\sqrt{82^2+55^2-2\times82\times55\times\cos76}$ oe OR **M1** for $82^2+55^2-2\times82\times55\times\cos76$ oe **A1** for 7570 or 7566 to 7567
7(b)	66.1 or 66.2 or 66.13 to 66.17	**3**	**M2** for $\frac{82\times\sin76}{\textit{their}\ \textbf{(a)}}$ oe or **M1** for $\frac{82}{\sin C}=\frac{\textit{their}\ \textbf{(a)}}{\sin76}$ oe
7(c)	13.3 or 13.30 to 13.31	**3**	**M2** for AG = 55 cos 76 oe or **M1** for recognition that CG is perpendicular to AB